

Predictive State Quotients and Irreducible Hidden-State Ambiguity in Reversible Dynamics

Exact finite-state observability, recurrence, and exhaustive lattice-gas classification

Scott A. Sundy

July 2026

Abstract

A repeated complete state in deterministic dynamics fixes the entire subsequent future, but a repeated observation need not. This study formulates exact future-equivalence for a finite deterministic system (X, F) observed through $h : X \rightarrow Y$, computes the resulting predictive partition by finite refinement, and applies the construction exhaustively to reversible lattice models. The formalism is positioned as a deterministic finite-state instance of established state-equivalence, automaton-minimization, bisimulation, observability, symbolic- dynamics, and predictive-state ideas rather than as a new general theory.

The principal computer-assisted result concerns the 3×3 periodic Hardy–Pomeau–de Pazzis square lattice gas in the complete four-particle, zero-momentum sector. All 9,153 microscopic states are enumerated. Sitewise density produces 495 present-observation classes; future-word refinement gives 6,948, 9,090, and then 9,126 stable predictive classes. Exactly 9,099 classes are singletons and 27 are doubletons. The 54 states in those doubletons decompose as

$$54 = 18 \times 3 = 9 \times 2 \times 3 :$$

18 disjoint collision-free microscopic cycles of least period three, paired into nine time-reversed orbit pairs, with three aligned phases per pair. Density is invariant under velocity reversal, and each exceptional period-three density word is invariant under temporal reversal up to cyclic phase. Consequently, the two microscopic cycles in each pair generate the same complete future density movie. An independent verifier reconstructs the sector, predictive partition, cycles, reversal pairing, and certificate from low-level HPP rules. Under a uniform prior, exactly one bit remains conditional on membership in the exceptional set, while the sector-wide residual conditional entropy is $54/9153 \approx 0.00590$ bits. The result is an exhaustive predictive-observability classification of this specified sector, not a claim about all lattice gases or physical dynamics.

1 Introduction

Determinism is a statement about complete states. If a complete state repeats, then its subsequent evolution repeats. Experimental and computational records, however, usually retain only an observation: a density field, image, aggregate, conserved quantity, or finite time series. Equality of such records is weaker than equality of the complete microscopic state.

The finite-state problem considered here is precise: for a deterministic map $F : X \rightarrow X$ and observation $h : X \rightarrow Y$, which complete states generate the same entire future observation sequence? This question sits within established lines of work. Indistinguishability of deterministic machine states under all future output words is central to Moore-machine and Nerode equivalence [1, 2]; partition refinement underlies deterministic minimization algorithms [13]; bisimulation formalizes behavioral equivalence of transition systems [3]; generating partitions and symbolic dynamics ask when observation itineraries distinguish trajectories [4]; observability theory asks when internal states are determined by outputs [5]; and predictive-state and computational-mechanics constructions represent dynamics through future measurement statistics [14, 6]. Logical reversibility of finite computation has a separate established literature [7]. Reversibility does not, by itself, make a coarse observation sufficient.

Accordingly, predictive partition refinement and future-equivalence are not claimed here as wholly new constructions. They are used as a common exact language for recurrence and observability in finite reversible systems. The original contribution is narrower:

1. an exhaustive predictive-observability classification of the stated 3×3 HPP sector;
2. identification of every density-indistinguishable microscopic state;
3. proof that the full exceptional set is exactly nine pairs of collision-free period-three microscopic cycles;
4. a general time-reversal criterion explaining why a velocity-blind observation makes those paired cycles future-equivalent; and
5. a reproducible computer-assisted certificate, checked by an independent low-level verifier, establishing completeness of the classification.

A second-order reversible cellular automaton gives the complementary outcome: two consecutive observations reconstruct every complete state. Additional finite and continuous examples distinguish exact complete-state recurrence, finite- horizon observational agreement, permanent future ambiguity, and near recurrence. None of these results implies that the physical universe is finite, exactly recurrent, or a lattice automaton.

2 Complete-state recurrence

Let X be a state space and $F : X \rightarrow X$ a deterministic discrete-time map. An orbit is $x_{t+1} = F(x_t)$.

Proposition 1 (Repeated complete state). *If $x_m = x_n$, then $x_{m+k} = x_{n+k}$ for every $k \geq 0$.*

Proof. For every $k \geq 0$,

$$x_{m+k} = F^k(x_m) = F^k(x_n) = x_{n+k}.$$

□

Reversibility is unnecessary. What matters is that the update is single-valued and that the repeated object is the complete state relative to that update law. Positions alone are incomplete for Newtonian mechanics; one time slice is incomplete for a second-order recurrence; and a collision model may require a hidden local memory variable.

If X is finite, every orbit is eventually periodic by the pigeonhole principle. If F is bijective, every orbit begins on a cycle because a bijection of a finite set is a permutation. These familiar facts concern complete states. They do not settle whether a reduced observation is predictive.

3 Observation closure and predictive equivalence

Let Y be an observation space and let $h : X \rightarrow Y$ be a measurement or coarse-graining map. The observed process is

$$y_t = h(F^t x_0).$$

A *complete state* is an element of X containing every variable required for the update F to be single-valued. An *observation* is the selected value $h(x)$; it need not be injective and should not be called physically indistinguishable without specifying h . Equal observations at one time, equal finite observation words, equal complete future observation sequences, and equal complete states are four distinct relations. The last is microscopic equality; the third is the predictive relation defined below.

Definition 1 (One-step closure). *The observation h is closed on X if there is a map $G : h(X) \rightarrow h(X)$ such that*

$$h \circ F = G \circ h.$$

Closure means that the present observed value alone determines the next observed value.

Theorem 1 (Exact closure criterion). *A deterministic reduced update G exists if and only if*

$$h(x) = h(x') \implies h(Fx) = h(Fx')$$

for all $x, x' \in X$.

Proof. Necessity follows from $h(Fx) = G(h(x))$. For sufficiency, define $G(y) = h(Fx)$ for any x satisfying $h(x) = y$. The stated implication makes this definition independent of the representative x . $\square$

When closure fails, a longer observation word may still determine the future. Define the length- r forward delay word

$$D_r(x) = (h(x), h(Fx), \dots, h(F^{r-1}x)).$$

Definition 2 (Future equivalence). *For $x, x' \in X$, write $x \sim_h x'$ when*

$$h(F^k x) = h(F^k x') \quad \text{for every } k \geq 0.$$

The equivalence classes are called predictive states for the observation h .

Theorem 2 (Minimal predictive quotient). *The relation $\sim_h$ is an equivalence relation preserved by F . The quotient $Q_h = X/\sim_h$ admits a deterministic update*

$$\overline{F}([x]) = [Fx]$$

and an observation decoder $\overline{h}([x]) = h(x)$. Moreover, Q_h is the unique coarsest deterministic state representation compatible with the complete future observation process.

Proof. Equivalence is immediate from equality of sequences. If $x \sim_h x'$, then the sequences beginning at Fx and Fx' are equal, so $Fx \sim_h Fx'$ and $\overline{F}$ is well defined. Likewise, equality at $k = 0$ makes $\overline{h}$ well defined.

Now let $p : X \rightarrow Q$ be any deterministic predictive representation with maps $T : Q \rightarrow Q$ and $o : Q \rightarrow Y$ satisfying $p \circ F = T \circ p$ and $h = o \circ p$. If $p(x) = p(x')$, then for every $k \geq 0$,

$$h(F^k x) = o(T^k p(x)) = o(T^k p(x')) = h(F^k x').$$

Thus p may merge only future-equivalent states. Its partition therefore refines $X/\sim_h$, proving that the future-equivalence quotient is coarsest. Any two coarsest quotients are isomorphic. $\square$

Corollary 1 (Reversible predictive dynamics). *If X is finite and F is bijective, then $\overline{F}$ is bijective on Q_h .*

Proof. The induced map is well defined. Since F is surjective, every quotient class has a predecessor class, so $\overline{F}$ is surjective. A surjection on a finite set is bijective. $\square$

A reduced observation can therefore fail to be closed while its exact predictive quotient remains reversible. Missing information is represented by the refinement of the raw observation into predictive classes.

4 Finite partition refinement

For finite X , the predictive quotient can be computed without selecting an arbitrary maximum time. Let $\mathcal{P}_1$ be the partition induced by h . Recursively, place x and x' in the same class of $\mathcal{P}_{r+1}$ exactly when

$$h(x) = h(x') \quad \text{and} \quad Fx \text{ and } Fx' \text{ lie in the same class of } \mathcal{P}_r.$$

Equivalently, $\mathcal{P}_r$ is the partition induced by D_r .

Theorem 3 (Finite stabilization). *For finite X , the refinement sequence*

$$\mathcal{P}_1 \preceq \mathcal{P}_2 \preceq \dots$$

stabilizes after finitely many strict refinements. Its stable partition is exactly $X/\sim_h$. If $q_1 = |h(X)|$, stabilization requires at most $|X| - q_1$ strict refinements.

Proof. Each strict refinement increases the number of classes by at least one, and no partition has more than $|X|$ classes. Hence at most $|X| - q_1$ strict refinements are possible. Two states share a class of $\mathcal{P}_r$ exactly when their first r observations agree. Sharing every refined class is therefore equivalent to equality of all future observations. At stabilization, successor-class refinement adds no distinction, so the stable relation is $\sim_h$. $\square$

This gives three exact outcomes:

- if $\mathcal{P}_1$ is stable, the raw observation is closed;
- if a stable partition is discrete, a finite delay word reconstructs every complete state;
- if a stable partition contains a nonsingleton class, no amount of future observation can distinguish the states in that class.

5 Exact reconstruction in a second-order cellular automaton

Consider a periodic binary lattice of length N with complete state $S_t = (x_{t-1}, x_t)$ and update

$$x_{t+1,i} = x_{t-1,i} \oplus x_{t,i-1} \oplus x_{t,i+1}.$$

The complete-state map $(x_{t-1}, x_t) \mapsto (x_t, x_{t+1})$ is bijective because

$$x_{t-1,i} = x_{t+1,i} \oplus x_{t,i-1} \oplus x_{t,i+1}.$$

Let the observation be the current configuration $h(S_t) = x_t$.

Theorem 4 (Two-snapshot reconstruction). *For every $N \geq 1$, one observed snapshot is insufficient, while two consecutive observed snapshots reconstruct the complete state exactly.*

Proof. There are 2^{2N} complete states but only 2^N current configurations, so one snapshot cannot be injective. Given (x_t, x_{t+1}) , the displayed inverse formula recovers x_{t-1} cell by cell. Thus $D_2(S_t) = (x_t, x_{t+1})$ is bijective. $\square$

The implementation exhaustively verifies the class-count sequence

$$(2^N, 2^{2N}, 2^{2N})$$

for $N = 2, \dots, 8$, covering 87,376 complete states. The repeated terminal count is the computational certificate that the partition has stabilized.

6 Time reversal and observation symmetry

Let $F : X \rightarrow X$ be bijective. A *time-reversal involution* is a map $R : X \rightarrow X$ satisfying

$$R^2 = I, \quad RFR = F^{-1}.$$

The second identity is equivalent to $RF^t = F^{-t}R$ for every integer t . The *orbit* of x is $\mathcal{O}(x) = \{F^t x : t \in \mathbb{Z}\}$, and its *reversed orbit* is $R\mathcal{O}(x) = \mathcal{O}(Rx)$. An observation is *reversal-invariant* when $h \circ R = h$.

Theorem 5 (Time-reversal observation identity). *Let F be bijective, let $R^2 = I$ and $RFR = F^{-1}$, and let $h(Rx) = h(x)$ for every $x \in X$. Then*

$$h(F^t Rx) = h(F^{-t}x)$$

for every $x \in X$ and every integer t .

Proof. The reversibility relation gives $F^t R = RF^{-t}$. Therefore

$$h(F^t Rx) = h(RF^{-t}x) = h(F^{-t}x),$$

where the last equality uses $h \circ R = h$. □

The theorem says that observing the forward evolution of the reversed state gives the original observation movie played backward. It does not yet say that the forward movies of x and Rx agree. That requires a symmetry of the observation word itself.

For a state of least period p , define its *density word*, or more generally its period- p observation word, by

$$w_h(x) = (h(x), h(Fx), \dots, h(F^{p-1}x)).$$

Two period- p words are *cyclically equivalent* when one is obtained from the other by a cyclic phase shift. A pair of states is *phase aligned* when the selected representatives begin at phases that make their future observation sequences agree term by term.

Theorem 6 (Periodic observation-word reversal criterion). *Assume the hypotheses of theorem 5. Let x have least period p . Suppose there is an integer s , understood modulo p , such that*

$$h(F^t x) = h(F^{s-t} x) \quad \text{for every } t \pmod{p}.$$

Then the phase-aligned state

$$z = F^{-s} Rx = RF^s x$$

on the reversed orbit satisfies

$$h(F^t z) = h(F^t x) \quad \text{for every } t \geq 0.$$

Thus x and z are future-equivalent under h .

Proof. For any integer t , theorem 5 and periodicity give

$$\begin{aligned} h(F^t z) &= h(F^{t-s} Rx) \\ &= h(F^{s-t} x) \\ &= h(F^t x). \end{aligned}$$

The last equality is exactly the assumed reversal symmetry of the period- p observation word. In particular it holds for every $t \geq 0$. □

Four cases should be kept separate.

1. If $R\mathcal{O}(x) = \mathcal{O}(x)$, time reversal maps the microscopic orbit to itself; z is then another phase of the same orbit, possibly x itself.
2. If $R\mathcal{O}(x) \neq \mathcal{O}(x)$, the reversed orbit is a distinct microscopic orbit. Distinctness alone says nothing about its observations.
3. The distinct orbits are observationally future-equivalent precisely for the aligned representatives covered by the criterion; their complete states remain unequal.
4. The unshifted state Rx need not agree with x at phase zero. The factor F^{-s} supplies the required phase alignment. Therefore “identical after a phase shift” is not the same statement as literal equality of the two displayed words without alignment.

7 Exhaustive density observability in the HPP lattice gas

7.1 HPP dynamics and the correct reversal map

The square Hardy–Pomeau–de Pazzis lattice gas [15] has four Boolean velocity channels at each site of a periodic square lattice. At a site containing exactly a head-on east–west pair, collision replaces that pair by north–south; conversely, an isolated north–south pair becomes east–west. All other local masks remain unchanged. Let C denote simultaneous collision and S one-site streaming. The project uses the update convention

$$F = S \circ C.$$

Both C and S are bijections, with $C^{-1} = C$.

Let V reverse every velocity channel, $N \leftrightarrow S$ and $E \leftrightarrow W$. Bare velocity reversal is not, for this collide-then-stream convention, the global time-reversal involution on arbitrary states. The correct map is

$$R = C \circ V.$$

Because C commutes with V , $R^2 = I$. Also $VSV = S^{-1}$, and therefore

$$RFR = (CV)(SC)(CV) = CVSV = CS^{-1} = F^{-1}.$$

The sitewise density observation is

$$h(x)_j = \sum_{d \in \{N, E, S, W\}} x_{j,d}.$$

Both collision and velocity reversal preserve the particle count at each site, so $h(Rx) = h(x)$. The hypotheses of theorem 5 therefore hold on the entire HPP state space.

A state is *collision-free* when no site contains an active isolated head-on pair. On such a state $Cx = x$, and on its velocity reversal $CVx = Vx$. Consequently the HPP reversal reduces to ordinary velocity reversal,

$$Rx = Vx,$$

throughout the exceptional set classified below. This is why the permanent ambiguity can be described physically as a velocity-reversal ambiguity without misstating the global reversal law.

7.2 Exhaustive sector and predictive partition

The complete state space examined here is the full set of microstates on a 3×3 torus satisfying:

- exactly four particles;
- total momentum $(0, 0)$;
- at most one particle per velocity channel, as required by the Boolean lattice gas.

This forward-invariant sector contains exactly 9,153 states. No sampling enters the classification.

A *predictive singleton* is a one-state class of $X/\sim_h$. A *predictive doubleton* is a two-state class. In the latter case, the two complete states are unequal but produce equal complete future density sequences. This is observational indistinguishability under the specified density map, not unqualified physical indistinguishability.

Observation-word length	1	2	3	4	5
Number of classes	495	6,948	9,090	9,126	9,126

Table 1: Exact partition refinement for sitewise density in the four-particle, zero-momentum 3×3 HPP sector. Equality of the final two counts certifies stabilization of the deterministic refinement.

The stable quotient has 9,126 classes: 9,099 predictive singletons and 27 predictive doubletons. Thus 9,099 of the 9,153 microscopic states are identified uniquely by their complete future density sequence, while exactly 54 states remain ambiguous. Their structure is

$$54 = 18 \times 3 = 9 \times 2 \times 3.$$

There are 18 microscopic cycles, three phases per cycle, nine time-reversal pairs, two distinct microscopic cycles per pair, and three aligned phase doubletons per pair. Equivalently,

$$27 = 9 \times 3$$

is the exact number of predictive doubletons.

7.3 HPP application of the reversal criterion

Theorem 7 (Complete HPP ambiguity classification). *In the stated sector, all 54 ambiguous states are collision-free and have least period three. They decompose into 18 pairwise-disjoint microscopic cycles. The 18 cycles form exactly nine pairs under R , which here equals velocity reversal. For each pair, the two cycles are distinct, their velocity assignments differ, and their three phases can be aligned so that the density observations agree at every future time. Each pair contributes exactly three predictive doubletons. No other nonsingleton predictive class or ambiguity mechanism occurs in the enumerated sector.*

Proof. The exhaustive refinement first identifies all 27 nonsingleton classes. Every one is a doubleton, so their union contains exactly 54 distinct states. Direct successor decomposition partitions that union into 18 disjoint cycles of three states. The certificate checks that every listed state has zero active collision sites, that $F^3x = x$, and that neither $Fx = x$ nor $F^2x = x$; hence the least period is three.

For each cycle $\mathcal{O}$, applying R to all three states produces exactly one other certified cycle $R\mathcal{O}$. No certified cycle is paired with itself, no two pairs overlap, and the operation is involutive, giving exactly

nine reversal pairs. Since every exceptional state is collision-free, this pairing is literally obtained by reversing all four particle velocities.

For each pair, the certificate records a phase shift $s \in \{0, 1\}$ for which the period-three density word satisfies

$$h(F^t x) = h(F^{s-t} x) \quad (t \bmod 3).$$

By theorem 6, x and $F^{-s} R x$ have identical complete future density sequences. Applying F to both states gives the same statement for the other two aligned phases. Thus each orbit pair supplies three doubletons and all nine pairs supply $9 \times 3 = 27$.

Completeness follows from the stable partition itself: the certificate accounts for every member of every nonsingleton class, while all remaining 9,099 classes are singletons. Therefore no further density ambiguity, whether caused by time reversal or by another mechanism, occurs in this sector. $\square$

The period is also transparent geometrically. With no active collision, each particle retains its velocity and streams by one lattice site per update. On a side-three periodic lattice, three steps in any cardinal direction return a particle to its initial site. Therefore $F^3 x = x$ for every exceptional state. The certificate's one- and two-step inequalities establish that three, rather than a divisor of three, is the least period.

The 18 cycles also form one symmetry family under translations of the 3×3 torus and square-lattice dihedral transformations. This supplementary classification is checked over all exceptional states, but the time-reversal pairing, not the spatial symmetry orbit, is the mechanism needed for permanent future-density equivalence.

7.4 Aligned representative and density movie

Using channel masks $N = 1$, $E = 2$, $S = 4$, and $W = 8$, the first certified aligned pair begins with

$$\begin{aligned} a_0 &= (0, 0, 0, 0, 2, 4, 0, 1, 8), \\ b_0 &= (0, 0, 0, 0, 4, 8, 0, 2, 1). \end{aligned}$$

The velocity reversal of a_0 is phase 1 of the b cycle, so the recorded shift is $s = 1$ and $b_0 = F^{-1} R a_0$. The shared density word is

$$A, A, B,$$

where

$$A = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

Its reversal A, B, A becomes A, A, B after one cyclic shift, exactly as required by theorem 6. The microscopic masks differ even though every aligned density field is equal.

Intuitively: “Density sees where particles are but not which way they are moving. For these special period-three trajectories, the forward and velocity-reversed microscopic movies cast the same density shadow forever.” The statement is specific to the selected observation and exceptional orbit structure. It is stronger than equality at one time or over a finite horizon because

$$h(F^t a_i) = h(F^t b_i) \quad \text{for every } t \geq 0$$

for each of the 27 aligned phase pairs.

Paired collision-free period-three HPP cycles with one shared density movie
Cycle B is the velocity-reversed partner of cycle A (with cyclic phase alignment).

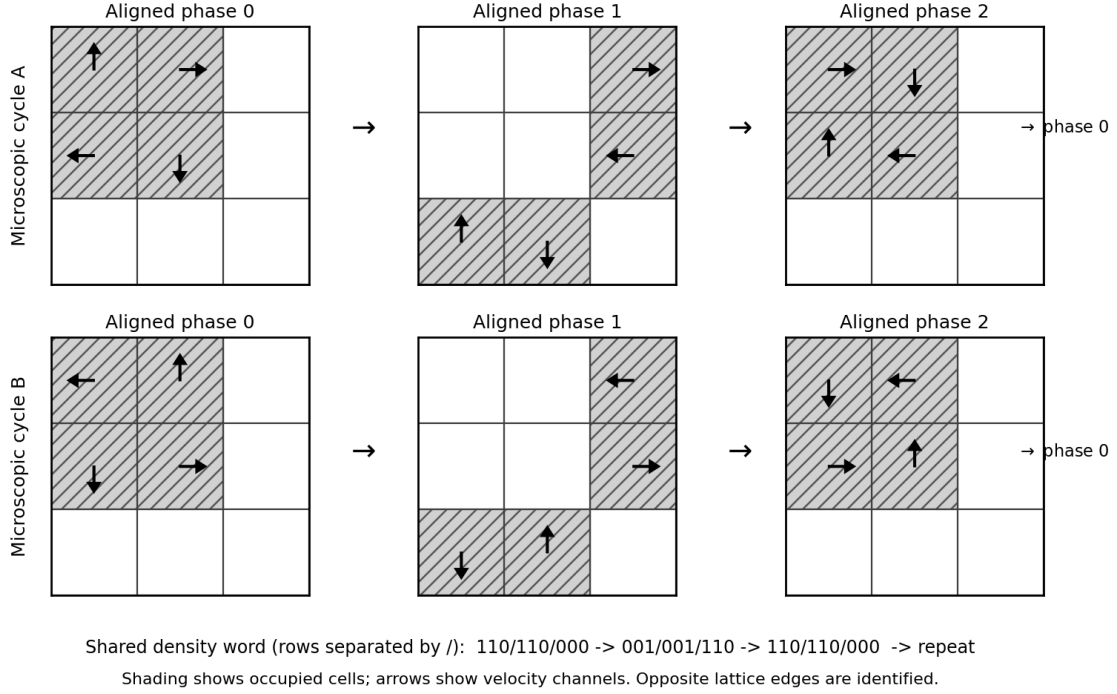


Figure 1: A self-contained representative of one certified pair on the 3×3 periodic lattice. The top and bottom rows show the three phases of microscopic cycles A and B. Velocity arrows distinguish the complete states; identical grayscale occupied-cell shading in each column shows the aligned density equality. Cycle B is the velocity-reversed partner of cycle A after the indicated phase alignment. Opposite lattice edges are identified, and the shared period-three density word is printed beneath the diagrams.

7.5 Computer-assisted proof and completeness certificate

The enumeration code discovers the stable predictive partition and writes the exact catalog. A separate verifier avoids the high-level classification routines: it reimplements channel collision, streaming, inverse streaming, state encoding, velocity reversal, and $R = C \circ V$ from low-level definitions. It then reads the committed exact-data files and independently asserts all of the following:

- exactly 9,153 sector states, each satisfying the four-particle and zero-momentum constraints;
- exactly 9,099 singleton classes and 27 doubletons in the stable predictive partition;
- exactly 54 distinct cataloged ambiguous states with no omission or duplicate assignment;
- exactly 18 disjoint cycles, each of least period three and collision-free;
- exactly nine involutive time-reversal pairings, with identical density at every recorded aligned phase;
- correct state, reversed-state, successor, predecessor, density, period, phase, pair, and collision fields in every certificate row; and

- predictive uniqueness of every nonexceptional state.

Assertions terminate with a nonzero status if any item fails. The catalog is thus a compact machine-checkable certificate, while the theorem explains the symmetry mechanism behind the certified exceptional set. Enumeration establishes that the list is complete; time-reversal symmetry explains why its members remain ambiguous forever.

7.6 Information-theoretic interpretation

Using Shannon conditional entropy [8], let X now denote a random microscopic state, uniformly distributed over the 9,153-state sector, let $W = \mathcal{F}_h(X)$ be its complete future density sequence, and let E be the event that X lies in the 54-state exceptional set. Every nonexceptional record has one compatible microscopic trajectory. Every exceptional record has exactly two, the two members of one predictive doubleton. Because the prior weights those two states equally,

$$H(X | W, E) = \log_2 2 = 1 \text{ bit.}$$

This is a conditional statement about the exceptional set; density does not generally lose one bit throughout the sector. Averaging over the uniform sector prior gives

$$H(X | W) = \frac{54}{9153} \log_2 2 \approx 0.00590 \text{ bits.}$$

The exceptional fraction is likewise

$$\frac{54}{9153} \approx 0.0058997 = 0.58997\%.$$

These values depend on the stated sector, observation map, and prior.

7.7 Reference orbit and ensemble studies

A separate specified 5×5 HPP microstate has exact period 175. At times 13 and 23 its density fields agree while the velocity microstates differ by eight bits and their next density fields differ. This demonstrates immediate failure of one-step density closure on a larger lattice.

Two deterministic paired ensembles test how hidden differences propagate:

- In 150 same-sector pairs, particle number, momentum, energy, and the full initial density field are identical. All density trajectories separate; 139 pairs show microscopic Hamming growth; and all 30 resolved cycle audits place the pair on different exact cycles.
- In 1,200 fixed-channel-filling pairs, all density trajectories separate; 1,158 show microscopic growth; 20 of 30 capped pair searches close for both states, and 19 of those 20 pairs have different periods.

8 A reversible hexagonal model with hidden collision memory

The six-direction model uses a local rotor bit to make selected collision rules exactly reversible. The rotor is part of the complete state even though it is not part of the particle-density observation. This is a designed reversible model, not the standard stochastic FHP collision rule of Frisch, Hasslacher, and Pomeau [16].

Across 900 paired trials, every pair begins with identical density, direction counts, momentum, particle number, and rotor field, but with a four-bit velocity reassignment. All 900 density trajectories separate. Particle damage grows in 852 pairs and hidden rotor fields separate in 849. The implementation also checks local collision involution, global inverse evolution, conservation laws, and 300 rotational-covariance trajectories.

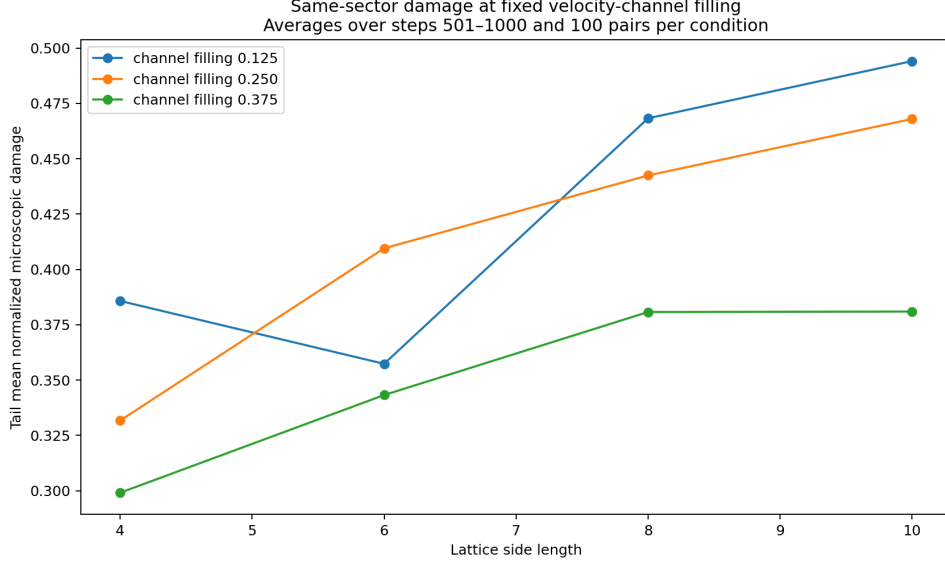


Figure 2: Mean normalized microscopic Hamming damage in the fixed-density HPP ensemble. The curves are finite-time damage diagnostics, not Lyapunov exponents. Every trajectory remains periodic because the state space is finite and the update is bijective.

9 Continuous exact and near recurrence

Finite-state periodicity must be separated from recurrence in continuous phase spaces.

9.1 Exact rational hard-particle orbit

An event-driven ring model uses equal masses, exact rational positions and velocities, and elastic velocity exchange. The selected labeled complete state returns after $T = 10$ with 36 collision events. Exchanging two initial velocity labels preserves the position tuple, velocity multiset, total momentum, and kinetic energy while changing the labeled future. Conserved quantities therefore do not constitute a complete state.

9.2 Irrational torus flow

Consider

$$q(t) = (t, \sqrt{2}t) \pmod{1}.$$

The flow is deterministic, reversible, compact, and measure preserving. An exact return at $T > 0$ would require both $T \in \mathbb{Z}$ and $\sqrt{2}T \in \mathbb{Z}$, which would make $\sqrt{2}$ rational. No exact positive return exists. Continued-fraction convergents nevertheless give arbitrarily small return errors, as shown in fig. 5. This is the distinction between exact point recurrence and Poincaré neighborhood recurrence [9, 10].

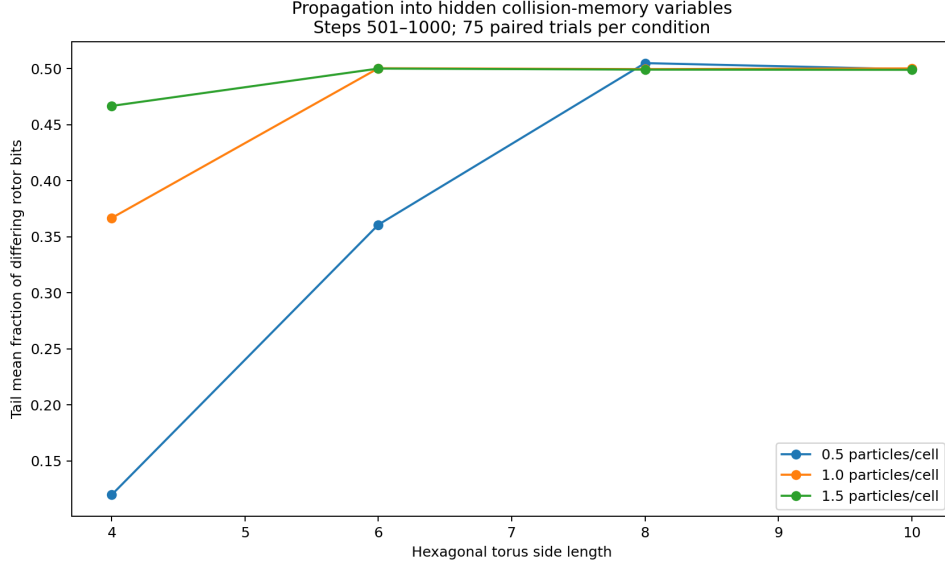


Figure 3: Propagation of microscopic differences into the hidden rotor field. Equality of density and conserved quantities does not fix the collision-memory state.

10 Quantum recurrence

A finite-dimensional Hilbert space is not a finite state set. Normalized vectors, and the associated projective pure states, form continua. For a time-independent Hamiltonian,

$$|\psi(t)\rangle = \sum_n c_n e^{-iE_n t/\hbar} |E_n\rangle.$$

Physical exact recurrence at time T requires equality up to global phase. Thus for every pair of occupied levels,

$$\frac{(E_n - E_m)T}{\hbar} \in 2\pi\mathbb{Z}.$$

Exact recurrence requires commensurate occupied energy gaps. With finitely many occupied levels, simultaneous phase approximation yields arbitrarily close returns, which underlies the quantum recurrence theorem [17, 18]. Continuous spectra, open-system decoherence, and infinite-dimensional limits require separate treatment.

The finite predictive-partition theorem in this paper does not apply directly to the continuum of quantum states. Its conceptual lesson still applies: equality of a reduced measurement record is not automatically equality of the quantum state, and exact recurrence must be defined at the level of physical states rather than raw vector phase.

11 Reproducibility

The repository distinguishes authoritative source files, deterministic scientific data, generated figures, and the compiled manuscript. The principal ambiguity catalog is independently checked rather than trusted because its row count looks plausible. A clean-reproduction profile regenerates deterministic outputs in a temporary tree and compares them against the committed data. Scientifically meaningful figure inputs are checked where binary PDF or rendering details are not guaranteed to be byte-for-byte stable.

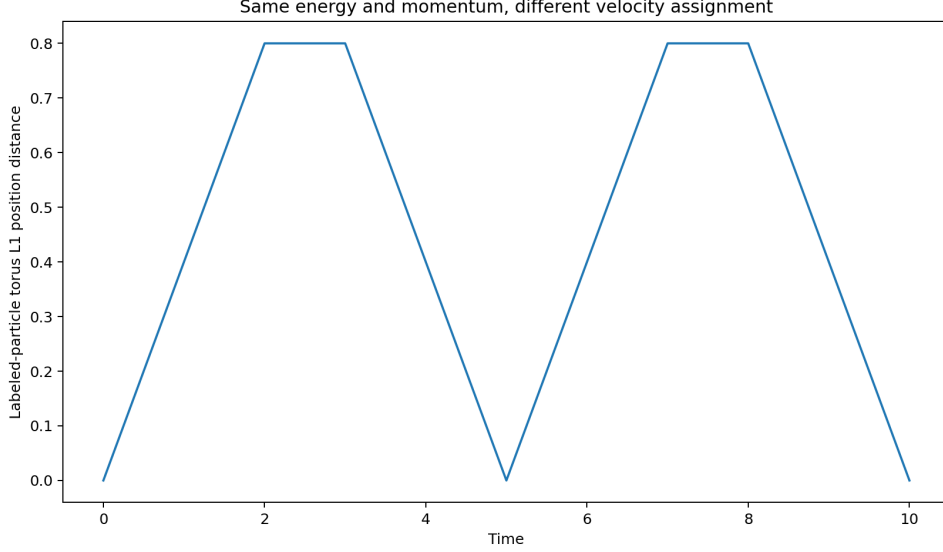


Figure 4: Exact position-space separation between two labeled hard-particle states with the same initial positions, velocity multiset, momentum, and kinetic energy.

The documented workflow from the repository root is:

```
python -m pip install -r requirements.txt -r requirements-dev.txt
python -m pip install -e .
python -m pytest
python scripts/verify_ambiguity_certificate.py
python scripts/validate.py reproduction
python scripts/validate.py data
python scripts/validate.py paper
python scripts/validate.py repository
python scripts/validate.py integrity
```

The ‘reproduce’ profile regenerates committed exact data and figures in place. The ‘all’ profile runs the tests, independent certificate verifier, clean reproduction, exact-data assertions, manuscript build, repository hygiene inspection, and integrity-manifest check. Every script uses repository-relative paths, exits nonzero on failure, and prints a concise successful result only after its assertions have passed. Declared cycle-search caps are recorded as censored rather than treated as evidence of nonrecurrence.

12 Scope and limitations

The exact predictive partition is computed for an explicitly finite deterministic system. The HPP result is exhaustive only for the stated 3×3 , four-particle, zero-momentum sector and sitewise-density observation; it is not a classification of other lattice sizes, particle sectors, collision rules, or coarse-grainings. The larger-lattice ensembles are finite computational studies with declared seeds, windows, and cycle caps. The square HPP model is anisotropic. The rotor model uses a deliberately introduced memory variable. The hard-particle example is integrable and commensurate by construction.

Moore/Nerode state equivalence, automaton minimization, bisimulation, symbolic coding, observability, and predictive-state representations supply the established conceptual setting. The contribution claimed here is the exhaustive HPP classification, the proof that its permanent ambiguity is exactly

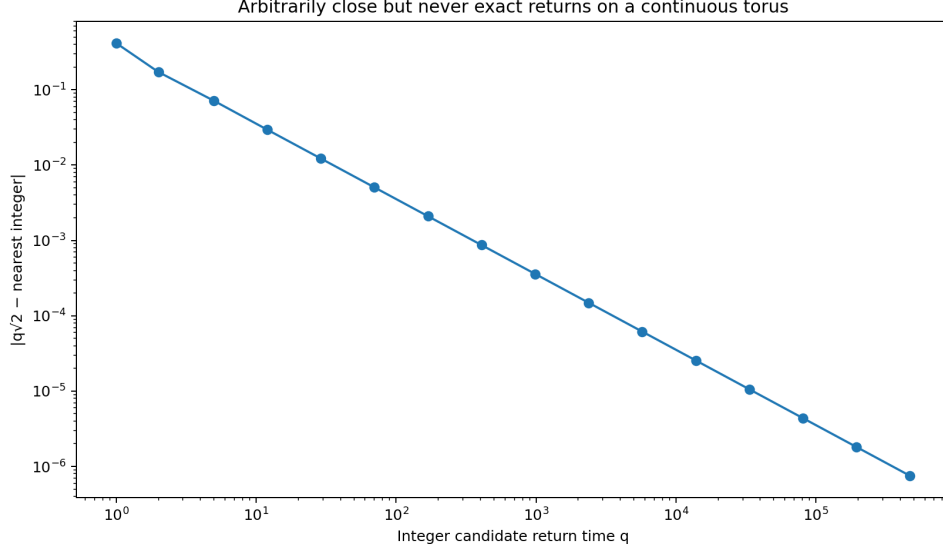


Figure 5: Near-return error for convergents to $\sqrt{2}$. The error decreases but never reaches zero.

nine paired period-three density movies, the precise time-reversal explanation, and the reproducible certificate of completeness. No claim is made of peer review, external replication, novelty priority for the general equivalence construction, or a universal physical law.

13 Conclusion

Repetition of a complete deterministic state fixes the future. Repetition of an observation does so only when the observation, possibly with a finite history, identifies the relevant predictive state. Finite partition refinement separates one-time equality, finite-horizon agreement, exact future equivalence, and microscopic equality without conflating them.

The HPP classification gives a complete nontrivial failure case. Of 9,153 states, 9,099 are uniquely determined by the full future density record. The remaining 54 states are not arbitrary exceptions: they are 18 collision-free least-period-three cycles, arranged as nine time-reversal pairs with three aligned phases. The reversal theorem explains why each pair has the same density future, and the independent verifier establishes that the catalog exhausts every nonsingleton predictive class. Under the uniform sector prior, the residual uncertainty is one bit conditional on the exceptional set and approximately 0.00590 bits when averaged over the whole sector.

References

- [1] E. F. Moore, “Gedanken-experiments on sequential machines,” in *Automata Studies*, Annals of Mathematics Studies Vol. 34, 129–153 (Princeton University Press, 1956).
- [2] A. Nerode, “Linear automaton transformations,” *Proceedings of the American Mathematical Society* **9**, 541–544 (1958). doi:10.1090/S0002-9939-1958-0135681-9.
- [3] D. Park, “Concurrency and automata on infinite sequences,” in *Theoretical Computer Science*, Lecture Notes in Computer Science Vol. 104, 167–183 (Springer, 1981). doi:10.1007/BFb0017309.

- [4] Ya. G. Sinai, “Markov partitions and C-diffeomorphisms,” *Functional Analysis and Its Applications* **2**, 61–82 (1968). doi:10.1007/BF01075361.
- [5] R. E. Kalman, “On the general theory of control systems,” in *Proceedings of the First International Congress of Automatic Control*, 481–492 (Butterworths, 1961).
- [6] M. L. Littman, R. S. Sutton, and S. Singh, “Predictive representations of state,” in *Advances in Neural Information Processing Systems 14*, 1555–1561 (MIT Press, 2002).
- [7] C. H. Bennett, “Logical reversibility of computation,” *IBM Journal of Research and Development* **17**, 525–532 (1973). doi:10.1147/rd.176.0525.
- [8] C. E. Shannon, “A mathematical theory of communication,” *Bell System Technical Journal* **27**, 379–423 and 623–656 (1948).
- [9] H. Poincaré, “Sur le problème des trois corps et les équations de la dynamique,” *Acta Mathematica* **13**, 1–270 (1890). doi:10.1007/BF02392506.
- [10] V. I. Arnold, *Mathematical Methods of Classical Mechanics*, 2nd ed. (Springer, 1989). doi:10.1007/978-1-4757-2063-1.
- [11] F. Takens, “Detecting strange attractors in turbulence,” in *Dynamical Systems and Turbulence, Warwick 1980*, Lecture Notes in Mathematics Vol. 898, 366–381 (Springer, 1981). doi:10.1007/BFb0091924.
- [12] T. Sauer, J. A. Yorke, and M. Casdagli, “Embedology,” *Journal of Statistical Physics* **65**, 579–616 (1991). doi:10.1007/BF01053745.
- [13] R. Paige and R. E. Tarjan, “Three partition refinement algorithms,” *SIAM Journal on Computing* **16**, 973–989 (1987). doi:10.1137/0216062.
- [14] J. P. Crutchfield and K. Young, “Inferring statistical complexity,” *Physical Review Letters* **63**, 105–108 (1989). doi:10.1103/PhysRevLett.63.105.
- [15] J. Hardy, Y. Pomeau, and O. de Pazzis, “Time evolution of a two-dimensional model system. I. Invariant states and time correlation functions,” *Journal of Mathematical Physics* **14**, 1746–1759 (1973). doi:10.1063/1.1666248.
- [16] U. Frisch, B. Hasslacher, and Y. Pomeau, “Lattice-gas automata for the Navier–Stokes equation,” *Physical Review Letters* **56**, 1505–1508 (1986). doi:10.1103/PhysRevLett.56.1505.
- [17] P. Bocchieri and A. Loinger, “Quantum recurrence theorem,” *Physical Review* **107**, 337–338 (1957). doi:10.1103/PhysRev.107.337.
- [18] L. S. Schulman, “Note on the quantum recurrence theorem,” *Physical Review A* **18**, 2379–2380 (1978). doi:10.1103/PhysRevA.18.2379.